

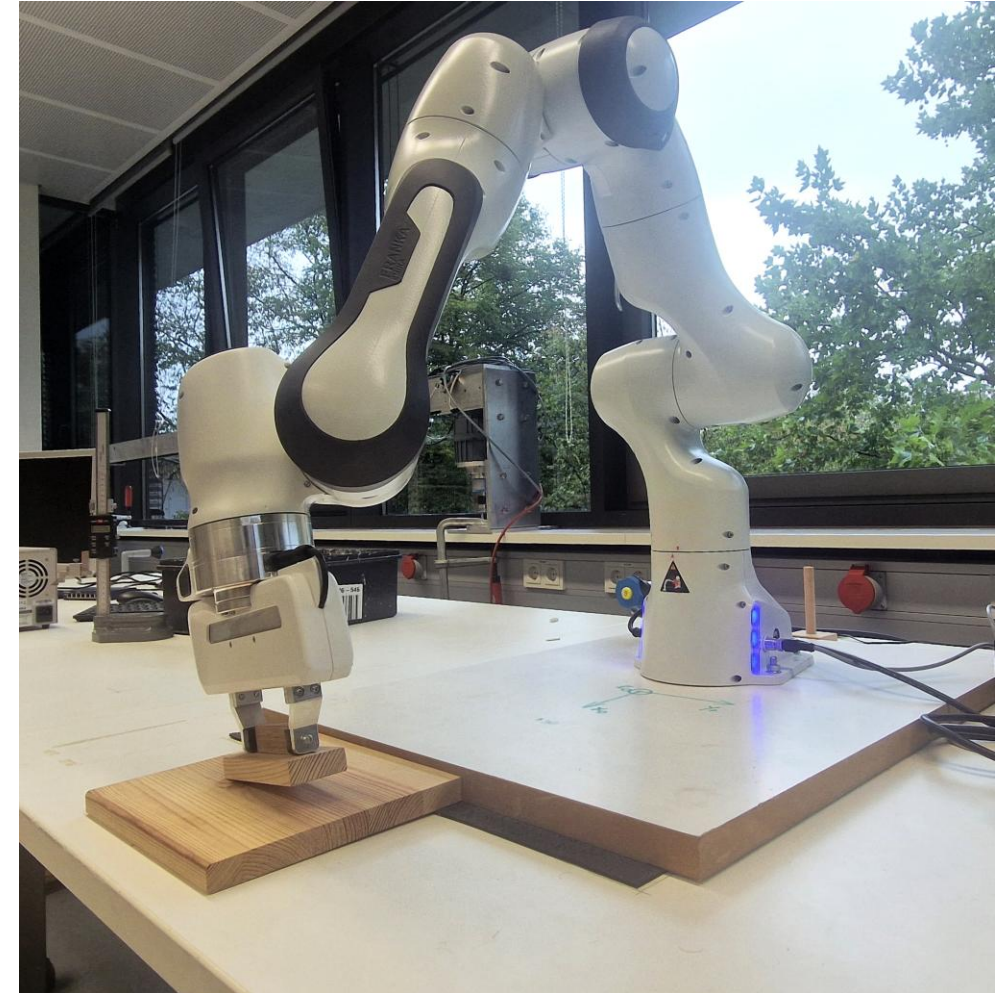
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Heykel Khadhraoui

Computational Engineering (M.Sc.)

Supervisors: Prof. Norbert Nitzsche · Prof. Ruth Otto

Date: 05.10.2026



Motivation

Requirement:

Grinding requires an aligned tool face.

Problem:

Orientation uncertainty after surface calibration

Joint friction → possible alignment blockage

Idea:

Cartesian impedance → compliant motion

Contact moment + compliant rotation
→ tool alignment



Overview

01 Theory

Grinding requirements and the alignment problem

02 Cartesian controller

Pose, compliance and real-time torque control

03 Contact experiments

Contact sequence and spring-response checks

04 Contact results

Stiffness and CoC effects on alignment

05 Null-space control and experiments

Control and measurement of extra joint motion

06 Conclusion and future work

Main findings and future work

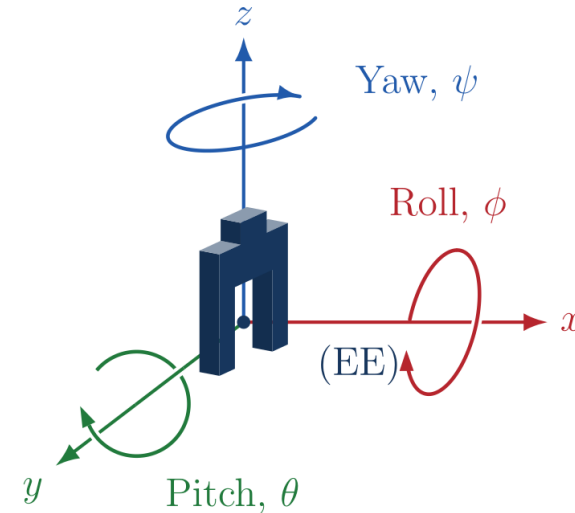
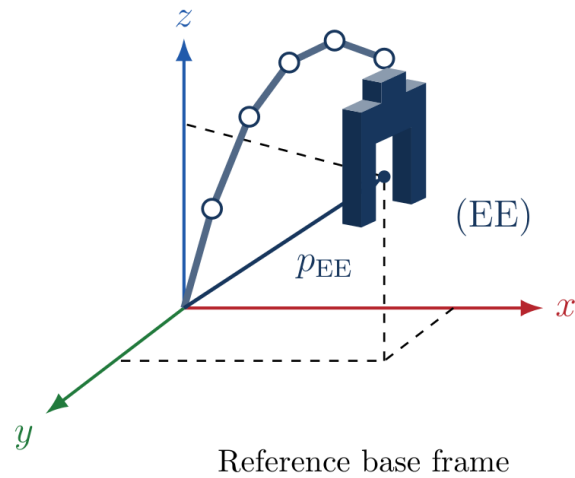
Cartesian pose

Cartesian pose of the end-effector (EE):

$$x_{EE} = [x \quad y \quad z \quad \varphi \quad \theta \quad \psi]^T$$

Position: 3 translations

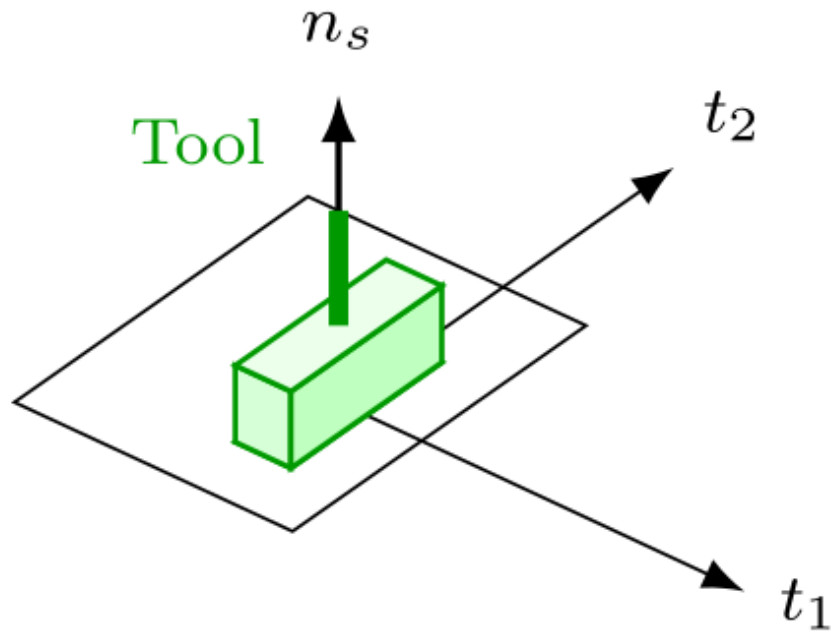
Orientation: 3 rotations



Forward kinematics: $x_{EE} = x_{EE}(q)$

Surface frame

Surface axes → force and moment components
and directions for impedance gains



n_s : upward surface normal

t_1, t_2 : perpendicular tangents

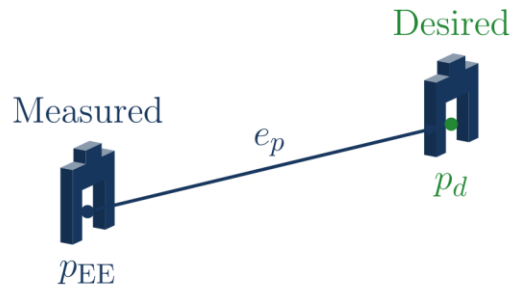
Cartesian impedance law

Force:

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

Positional error:

$$e_p = p_d - p_{EE}$$

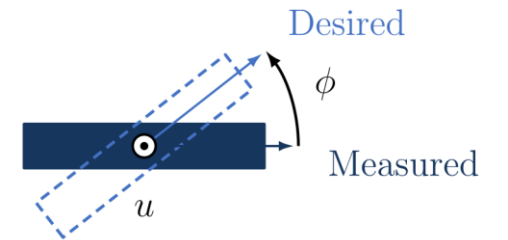


Moment:

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

Rotational error:

$$e_R = \phi u$$



Cartesian wrench:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

Surface frame: diagonal K and $D \rightarrow$ decoupled translation and rotation

Real-time impedance control loop

Commanded joint torques (motors):

$$\tau = J^T(q)F$$

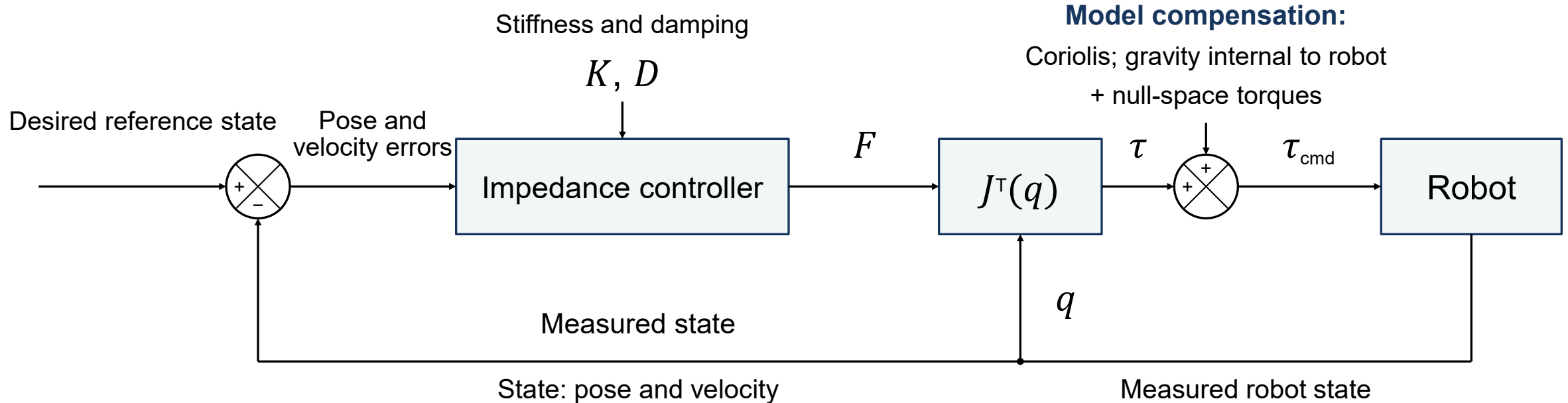
Cartesian contribution

Jacobian:

$$\dot{x}_{EE} = J(q)\dot{q}$$

Linear and angular velocity (geometric)

Real-time control: 1 ms cycle



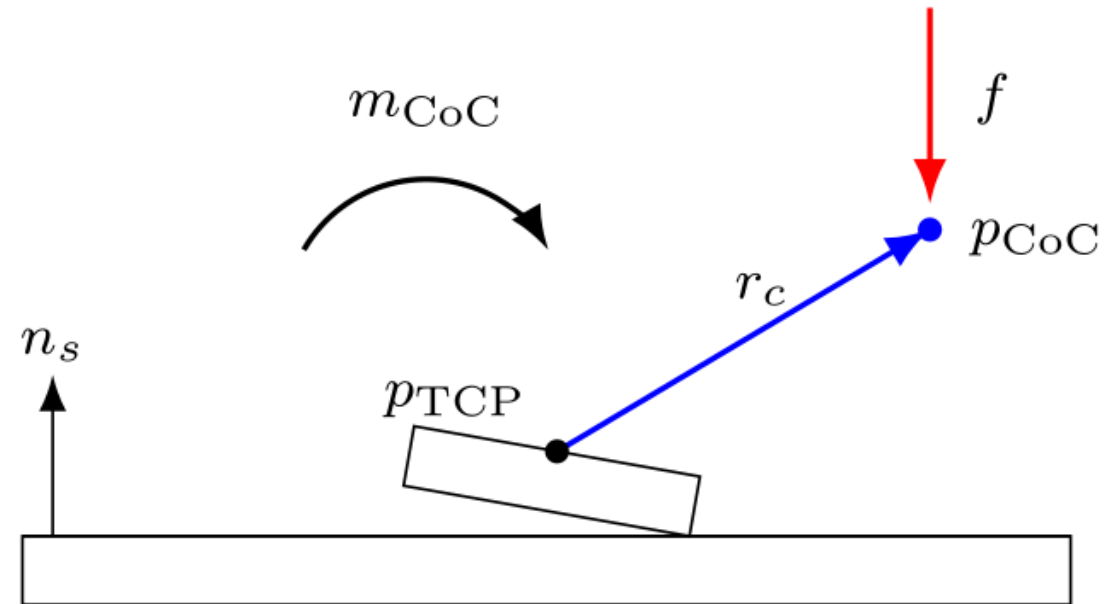
Centre of compliance: a virtual reference

Tool centre point (TCP):

Controlled point on the end-effector

Centre of compliance (CoC):

Virtual reference for
Cartesian stiffness and damping



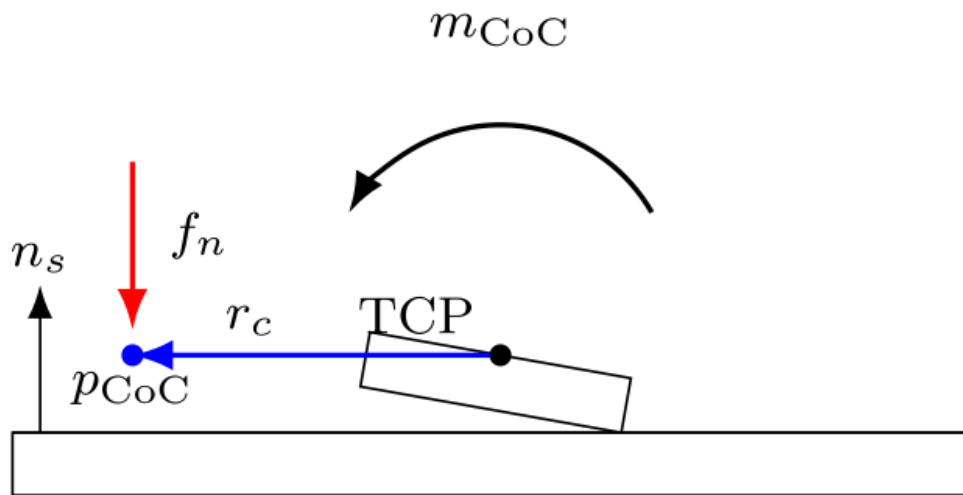
Equivalent force applied at the virtual CoC

Additional moment about the TCP:

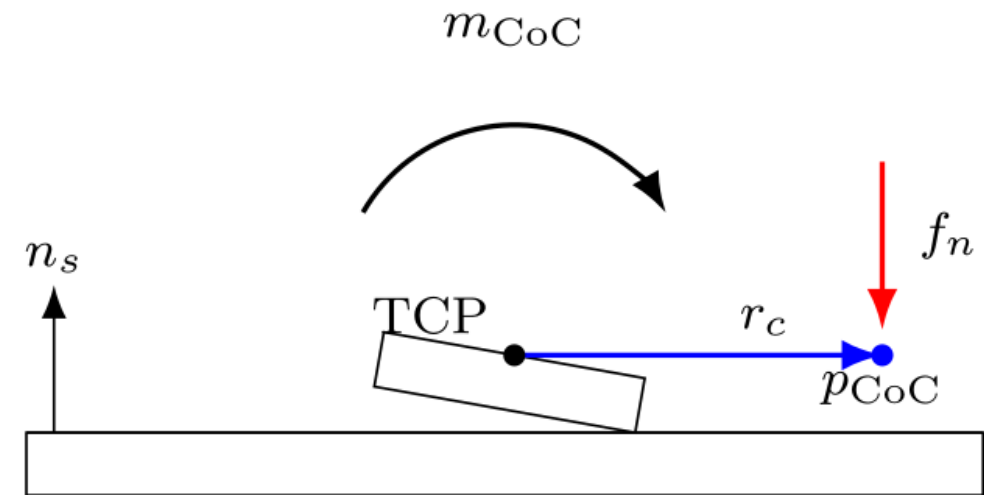
$$m_{CoC} = r_c \times f$$

Supporting and opposing CoC moments

Shift direction → supporting or opposing moment



(a) Supporting moment



(b) Opposing moment

Curved arrows: moment about TCP from the force at the virtual CoC

Translation–rotation coupling

Virtual CoC shift $\rightarrow$ coupled force and rotation

$$r_c = p_{\text{CoC}} - p_{\text{TCP}} \quad \text{Displacement from TCP to CoC}$$

$$A = \text{Ad}(r_c)$$

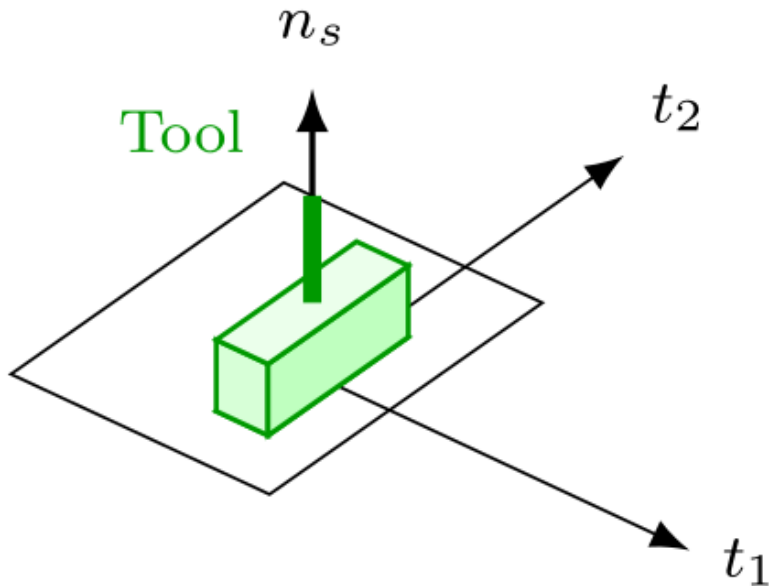
$$F_{\text{TCP}} = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{A^\top \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix} A}_{K_{\text{TCP}}} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{A^\top \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix} A}_{D_{\text{TCP}}} \begin{bmatrix} \dot{p}_d - \dot{p}_{\text{EE}} \\ \omega_d - \omega_{\text{EE}} \end{bmatrix}$$

Non-zero off-diagonal blocks in K and $D \rightarrow$ translation–rotation coupling

Contact phases and stiffness settings

Experiment phases:

1 Orient tool → 2 Approach surface → 3 Establish contact → 4 Grind



Compliant contact settings:

Normal translation (compliant):

$$K_{p,n} = 350 \text{ N/m}$$

Tangential translation (stiff):

$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

Rotation about the normal (stiff):

$$K_{R,n} = 50 \text{ N m/rad}$$

Rotation about the tangents (compliant):

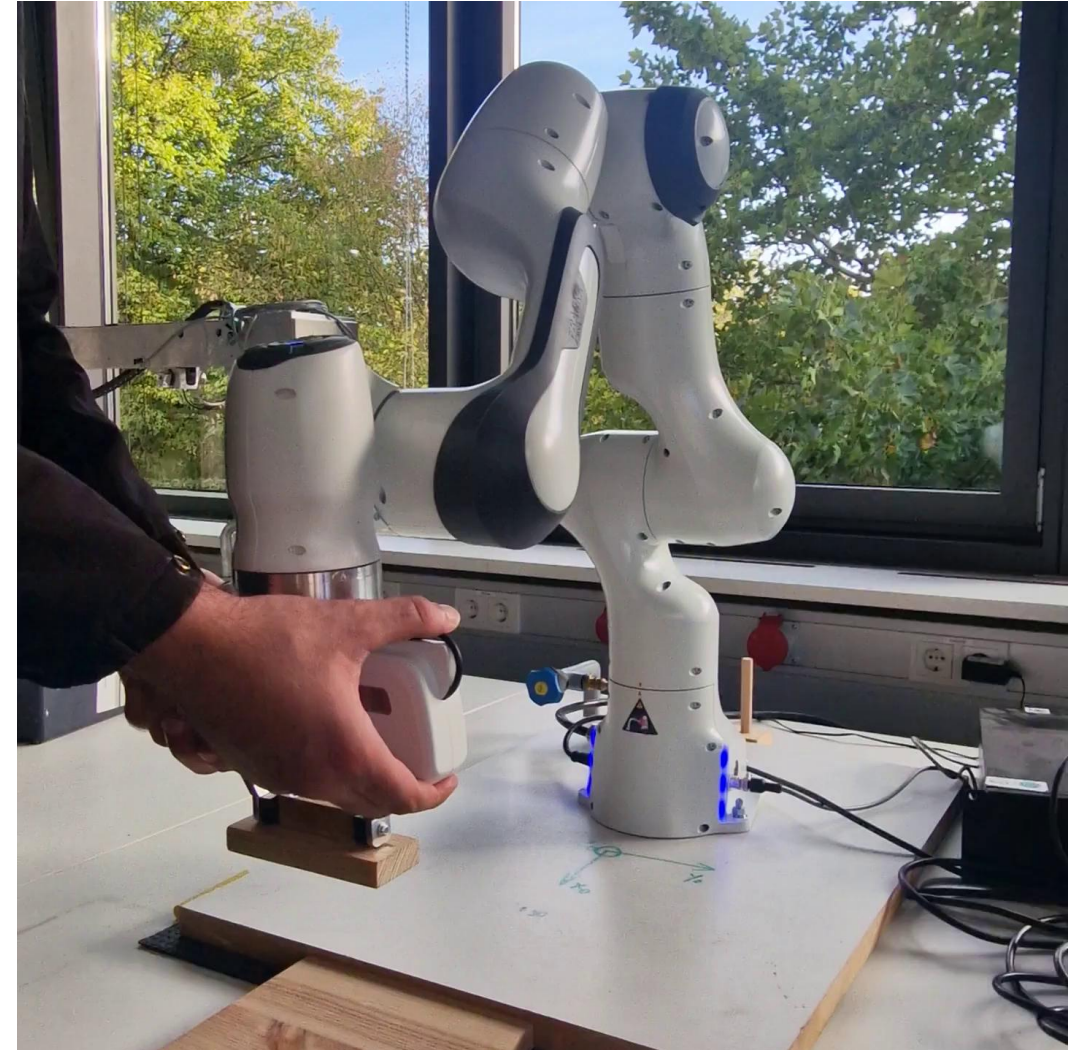
$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$

Impedance response to manual displacement

Spring-response check

Push the tool → force
Turn the tool → moment

Comparison: commanded, estimated
and quasi-static values



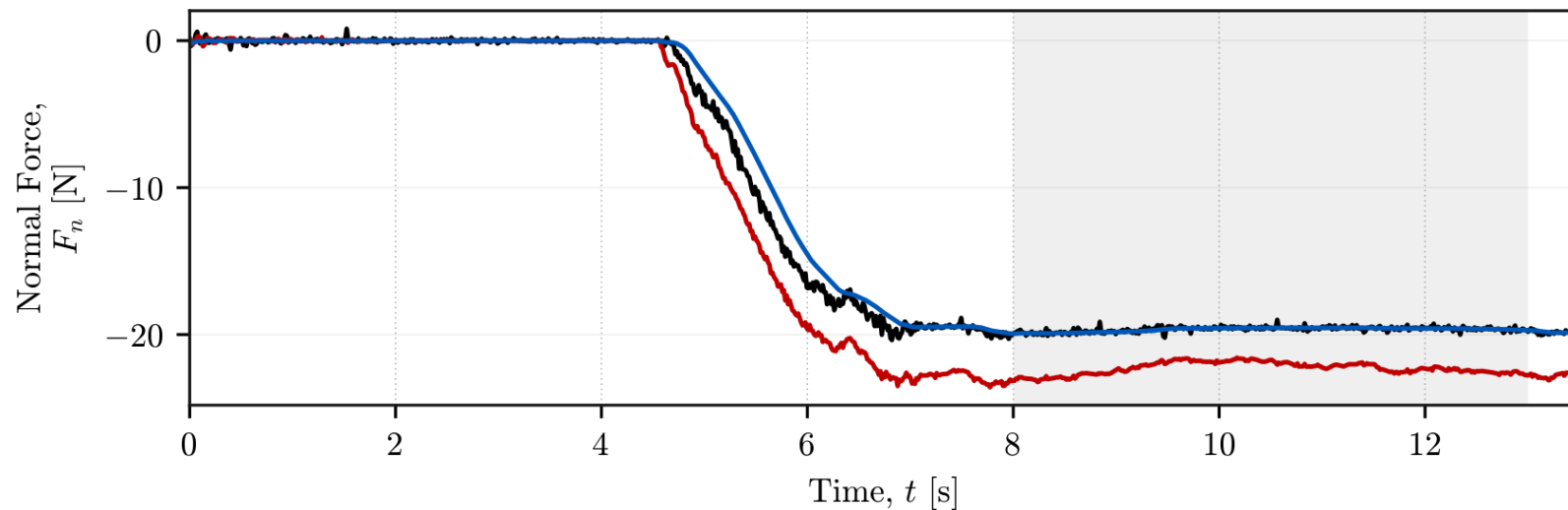
Normal-force plausibility assessment

Manual push: about 20 mm

Quasi-static calculation:

$$F_{n,qs} = K_{p,n} n_s^T (e_p - e_{p,0})$$

$$= (1000 \text{ N/m}) \cdot (-19.65 \cdot 10^{-3} \text{ m}) = -19.65 \text{ N}$$



— Commanded

— Model-estimated

— Quasi-static prediction

Quasi-static mean:
-19.65 N

Commanded mean:
-19.64 N

Model-estimated mean:
-22.29 N

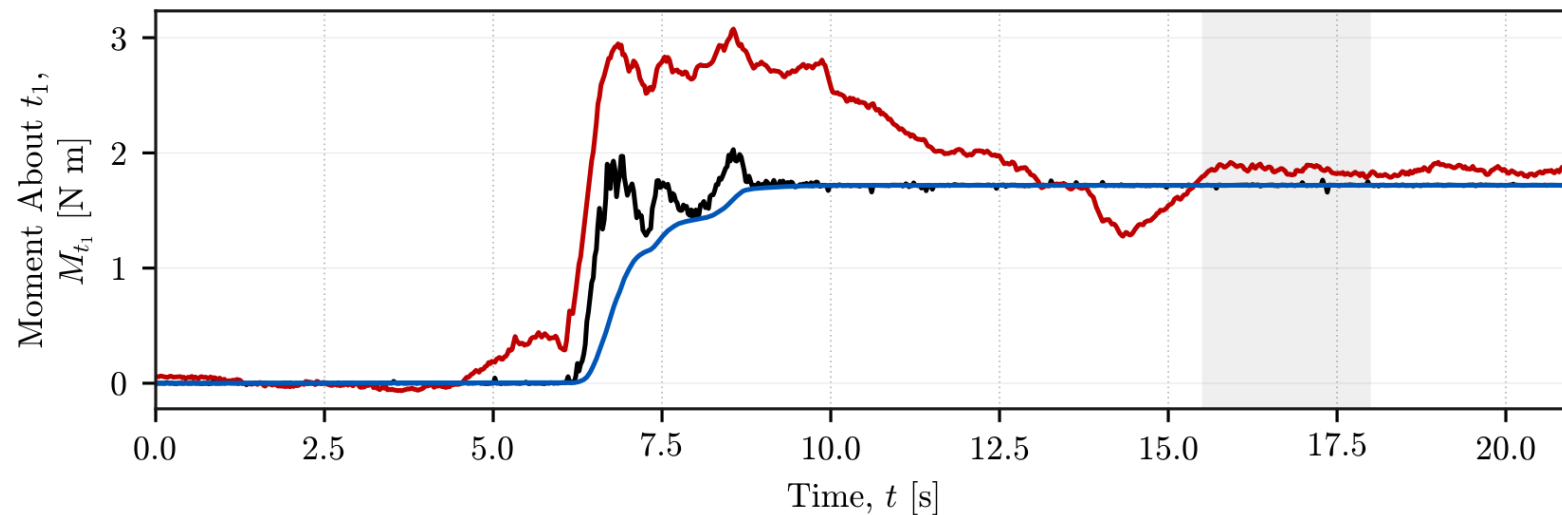
Moment plausibility assessment

Manual rotation: about 6.5°

Quasi-static calculation:

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$

$$= (15 \text{ N.m/rad}) \cdot \left(6.57 \cdot \frac{\pi}{180} \text{ rad} \right) \approx 1.72 \text{ N.m}$$



— Commanded

— Model-estimated

— Quasi-static prediction

Quasi-static mean:

1.72 N.m

Commanded mean:

1.72 N.m

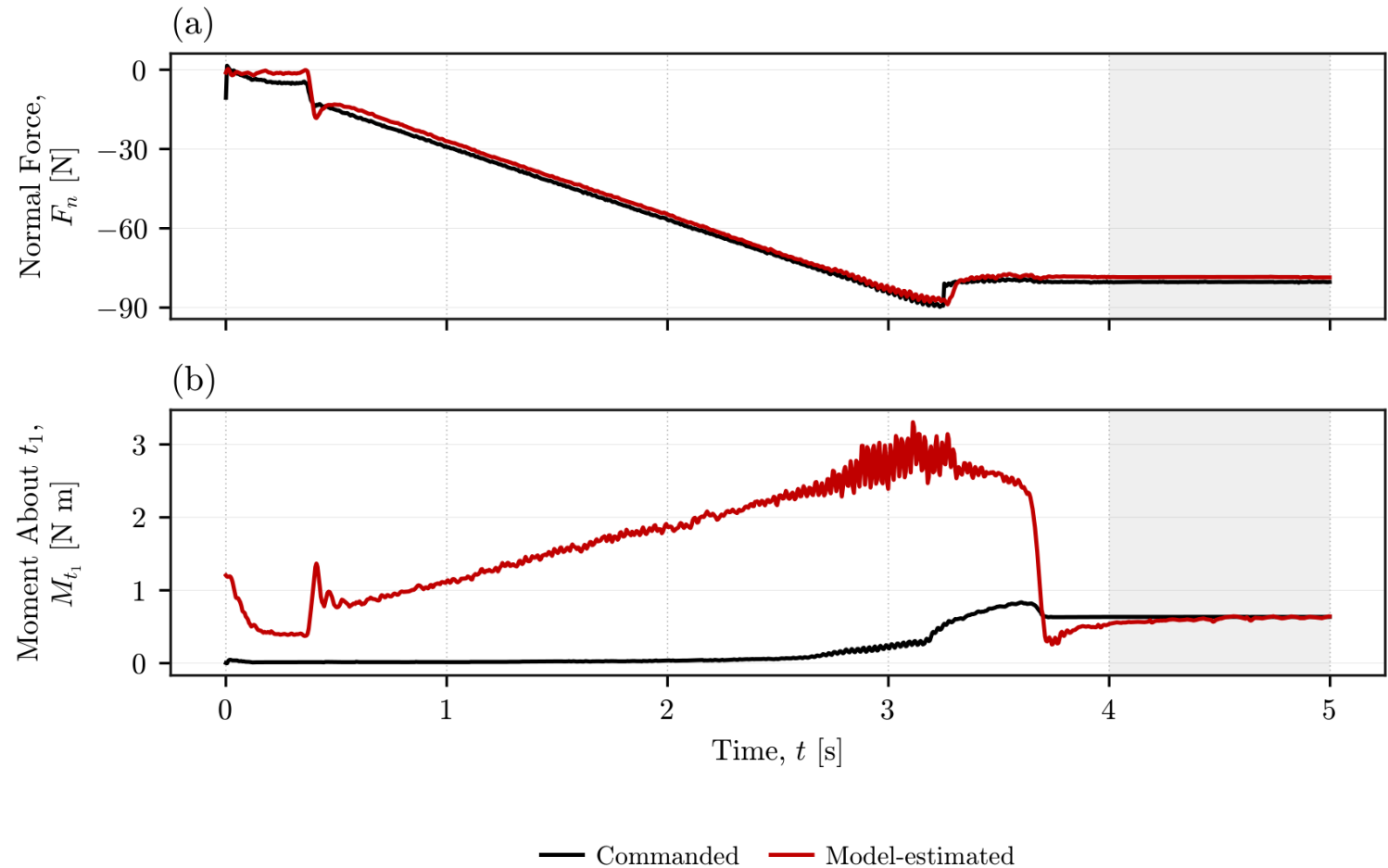
Model-estimated mean:

1.85 N.m

Force and moment during contact alignment

Force curves remain close

Larger estimated moment during alignment

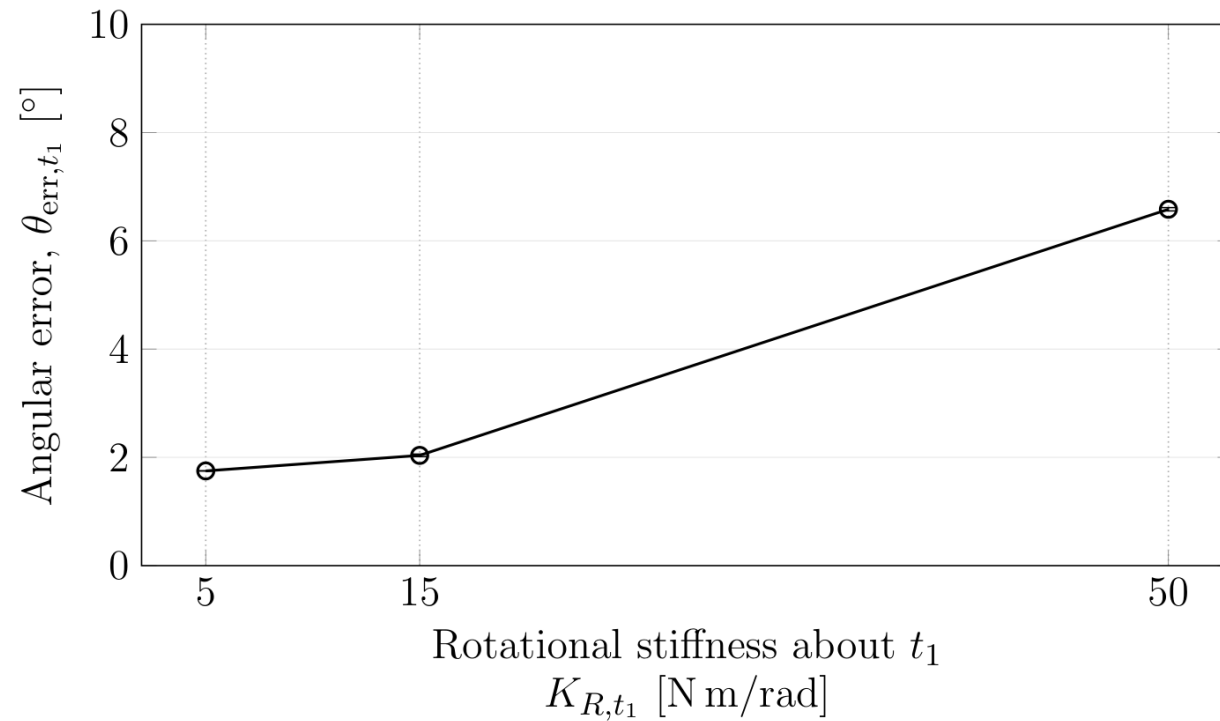


Rotational stiffness and angular error

Resistance to rotation about t_1 :

More stiffness
→ **more remaining tilt**

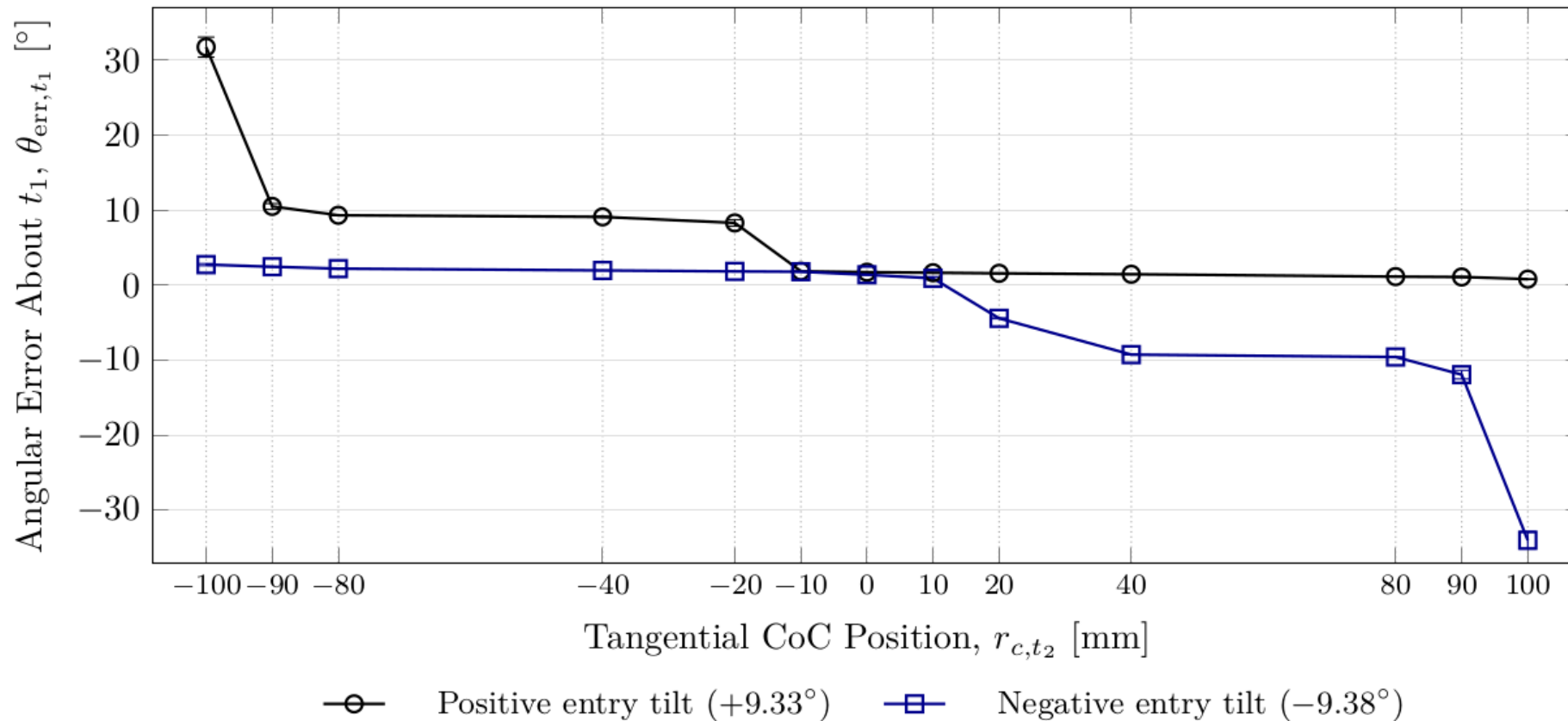
Reduced rotation from contact



—○— Measured angular offset
 $\theta_{meas,t_1} = (9.31-9.32)^\circ$

Effect of CoC position on angular error

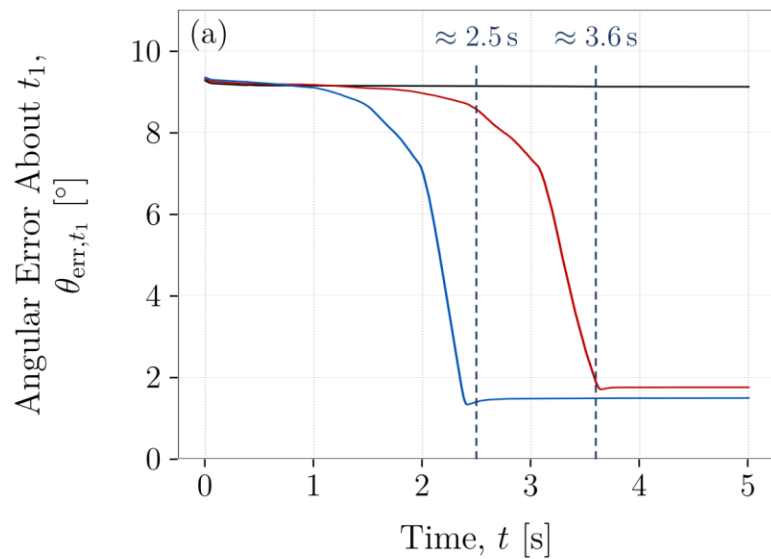
CoC displacement along t_2 : r_{c,t_2}



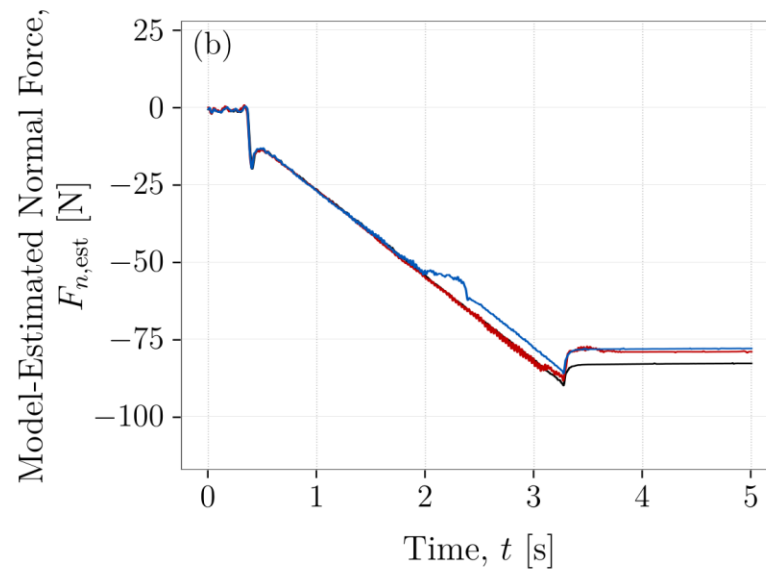
Larger shift $\neq$ better alignment · Useful direction depends on entry tilt

Contact response at different CoC positions

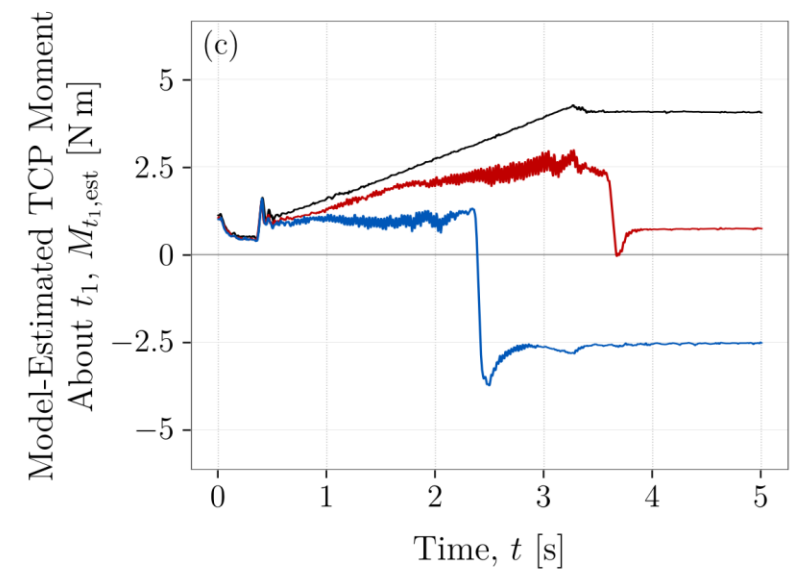
Angular error:



Normal force:



TCP moment:



— $r_{c,t_2} = -40$ mm
 — TCP: $r_{c,t_2} = 0$
 — $r_{c,t_2} = +40$ mm

Supporting CoC shift → faster turning · Similar normal forces

Null-space control: damping and conditioning

Extra freedom for arm posture

Arm posture changes $\rightarrow$ end-effector pose unchanged

Null space dimension = $7 - 6 = 1$ $\longrightarrow$ Null-space condition: $J(q) \dot{q}_{\text{null}} = 0$

Damping torque τ_d :

$\rightarrow$ Opposes null-space velocity

Less unwanted motion

Conditioning torque τ_σ :

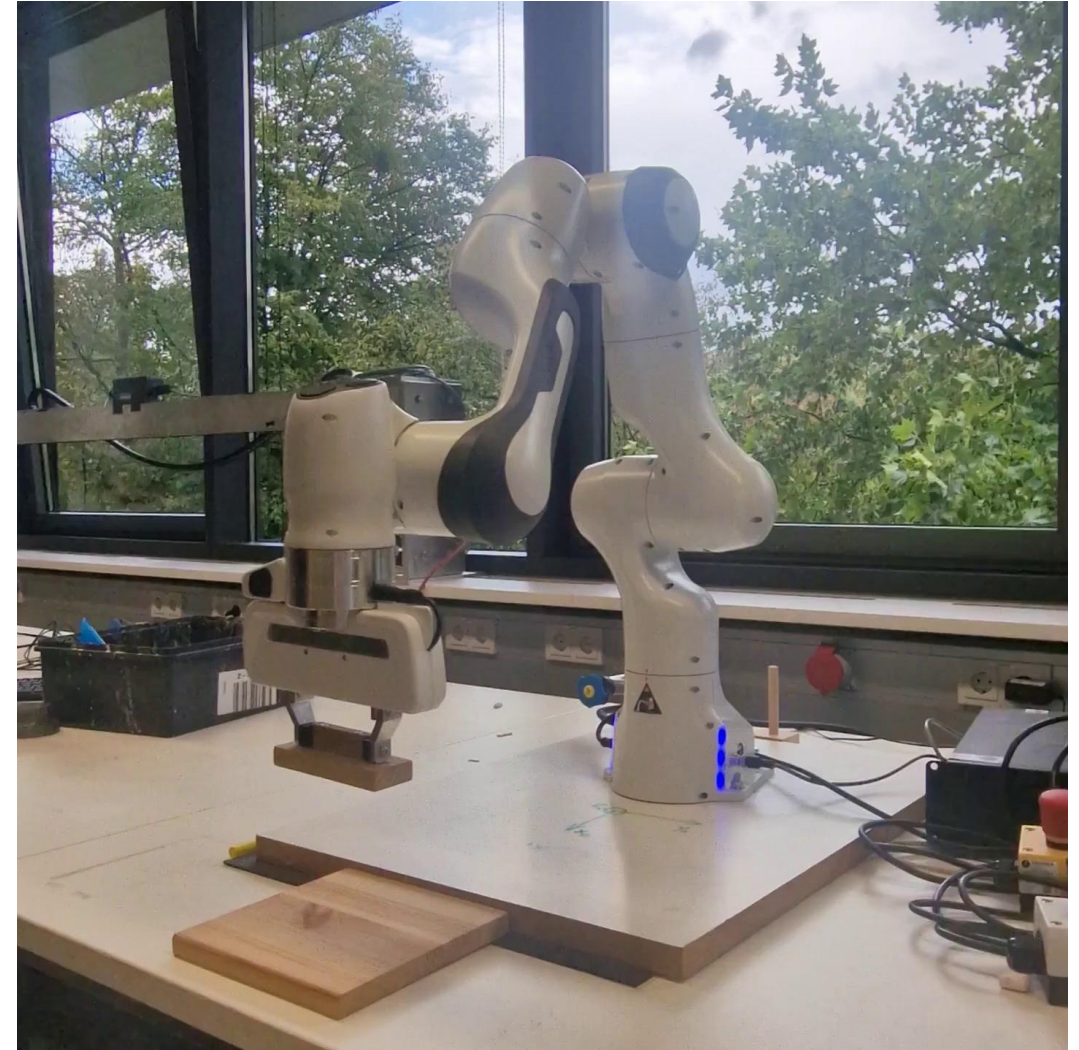
$\rightarrow$ Increase $\sigma_{\min}$ of J
(minimum singular value)
 $\rightarrow$ move away from singularities

Null-space damping demonstration

**Null-space control OFF →
Damping ON**

Damping → less extra arm motion

Tool pose held by the main controller

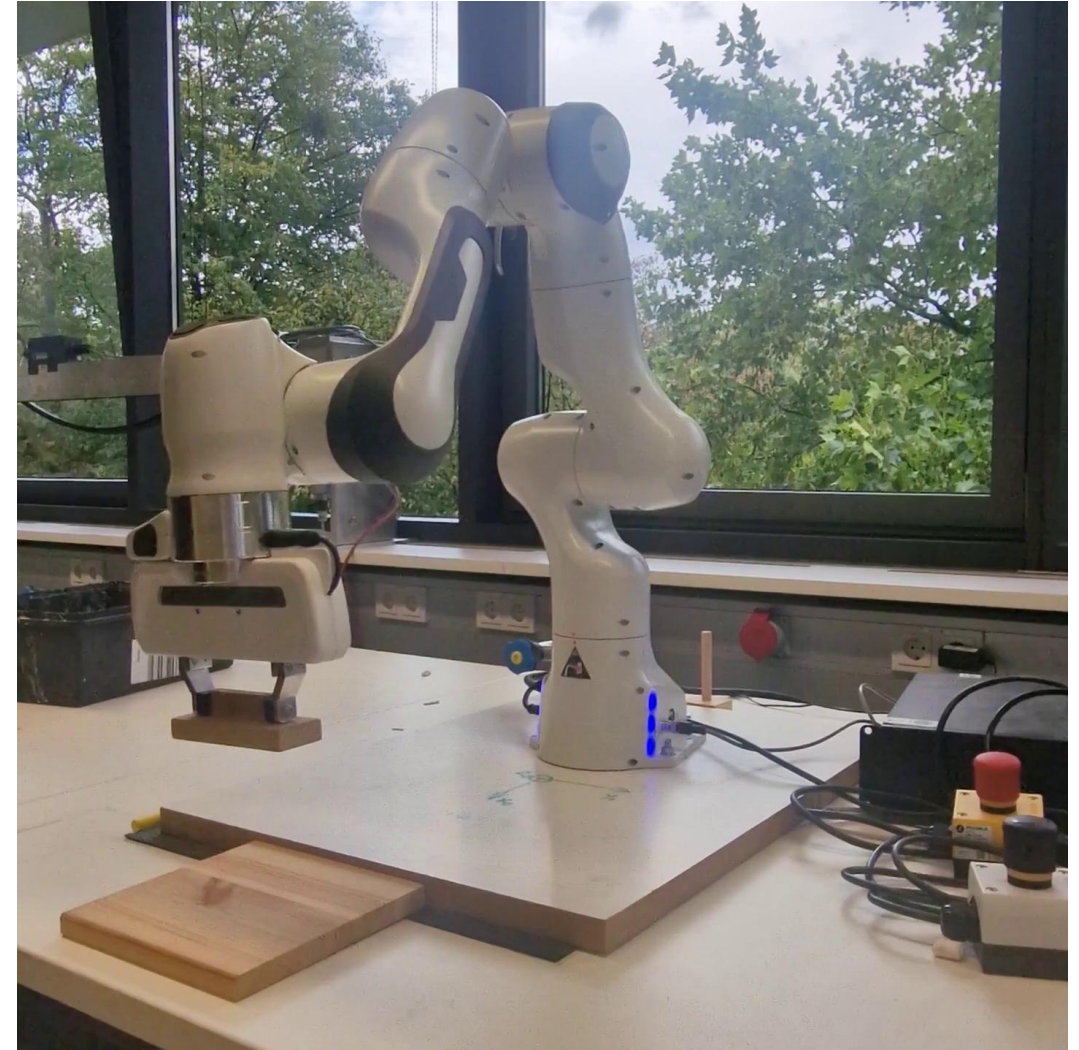


Null-space conditioning demonstration

Conditioning ON

Target: better arm posture

Higher minimum singular value
with unchanged tool pose



Null-space experiment: setup and conditions

Fixed tool pose + virtual disturbance force

Captured position and orientation held constant

$$\tau_{\text{dist}} = J(q)^{\top} f_{\text{dist}}$$

Virtual 20 N point force on link 3
→ equivalent joint torques

Joint 1 shown: largest peak disturbance torque

Four conditions · Three trials each

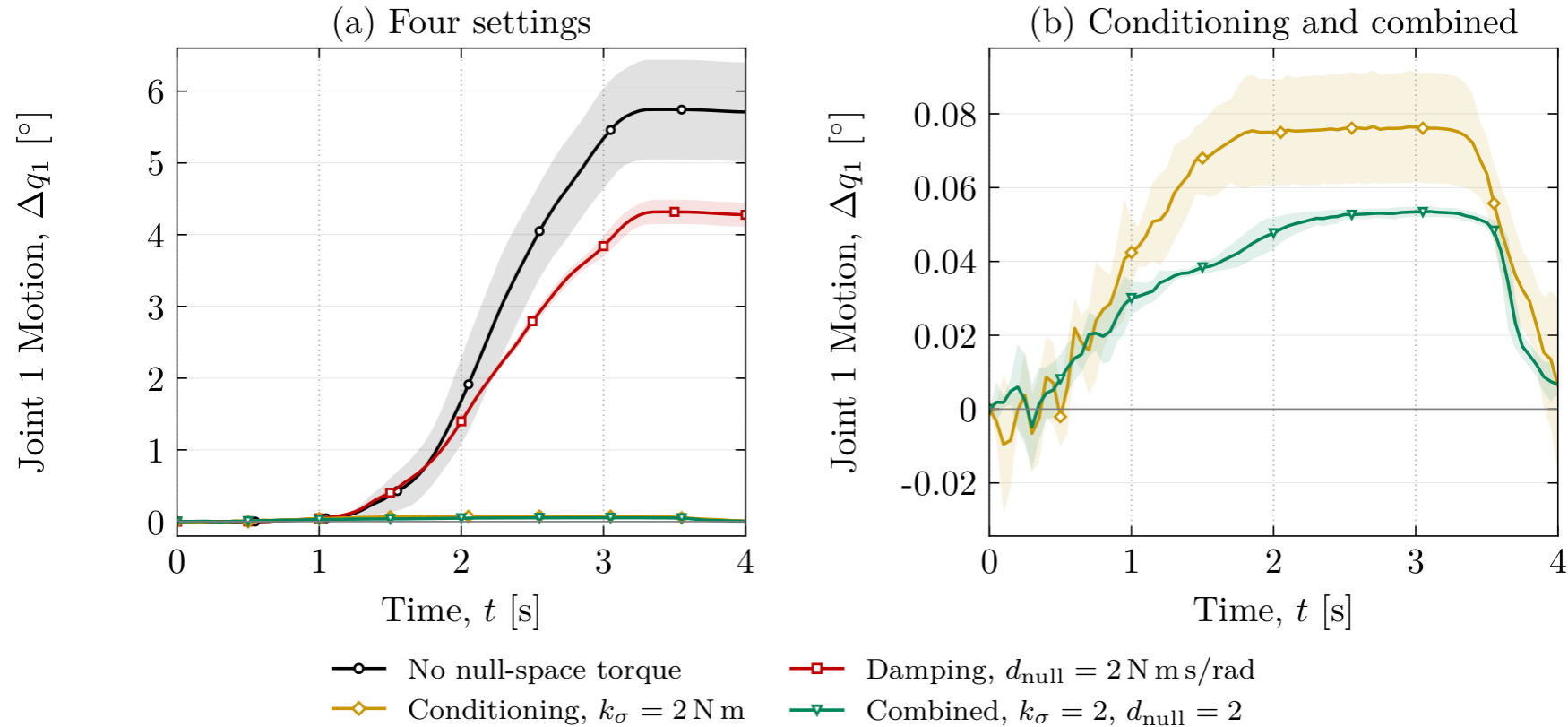
No null-space control torque

Damping: $d_{\text{null}} = 2 \text{ N m s/rad}$

Conditioning: $k_{\sigma} = 2 \text{ N m}$

Combined: conditioning + damping
 $k_{\sigma} = 2 \text{ N m}$ and $d_{\text{null}} = 2 \text{ N m s/rad}$

Joint 1 motion under null-space control

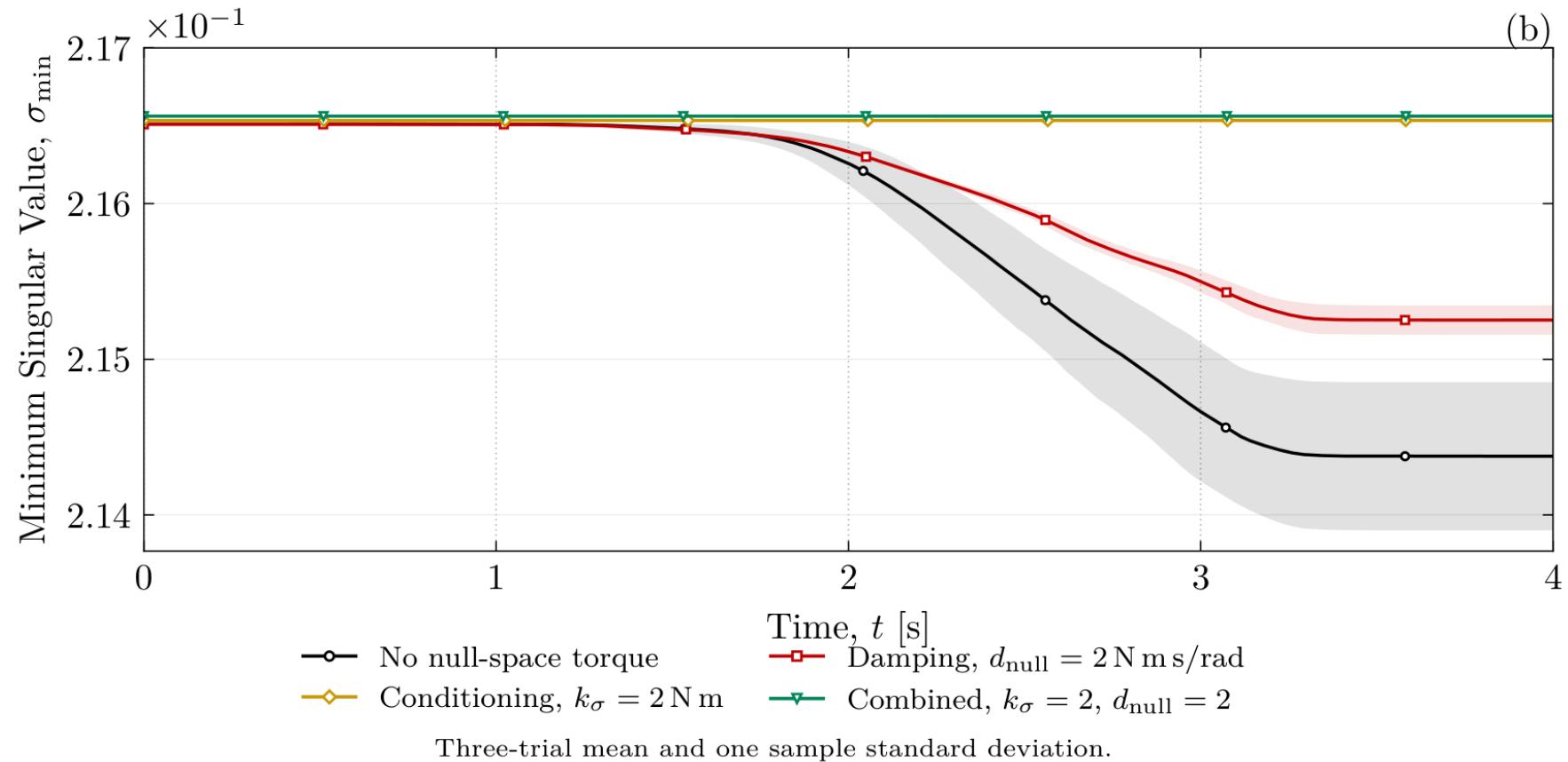


Three-trial mean and one sample standard deviation.

Conditioning (2 N m): repeated joint-motion reversals

Added damping: less total motion, remaining reversals

Jacobian conditioning: minimum singular value



Decreasing $\sigma_{\min}$: no null-space torque / damping alone

Nearly constant $\sigma_{\min}$: conditioning / combined control

Conclusion and future work

Contact → tool alignment · Null-space control → less arm motion

Main findings

- 1 Compliant rotation → reduced angular error
- 2 Higher rotational stiffness → larger angular error
- 3 CoC shift → supporting or opposing moment
- 4 Null-space control → less redundant motion

Future work:

More rigid tool mount
→ better measurements

Adaptive CoC:
supporting moment
Return to TCP after alignment

Opposing CoC moment during contact

Opposing moment

Opposing shift → hindered alignment

Comparison with the supporting CoC case



Null-space torques and projector

Null-space projector:

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Secondary joint torques $\rightarrow$ kinematic null space

J : joint $\rightarrow$ (EE) velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

Damping torque:

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity

Zero at zero joint velocity

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

Conditioning torque:

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

Seeks better local conditioning

Can act at zero joint velocity

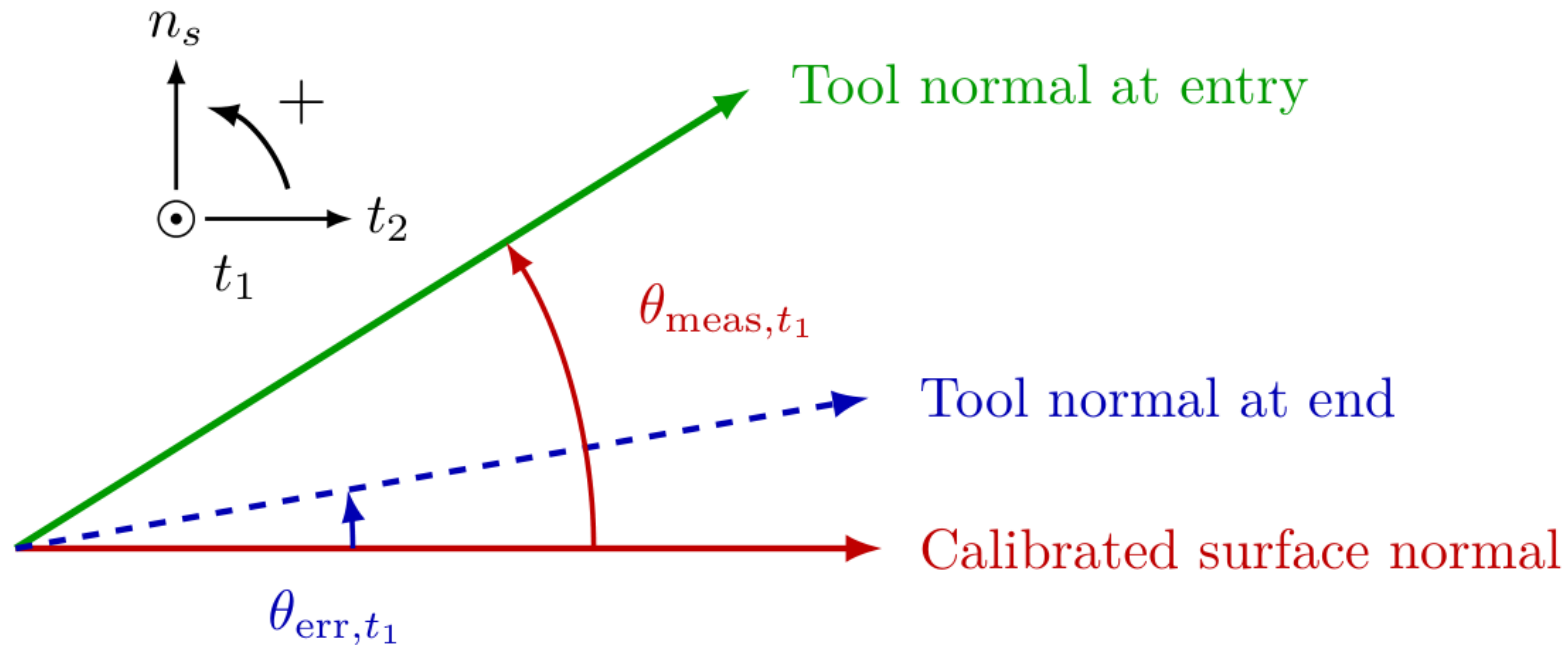
k_σ : torque magnitude [N m]

v_7 : unit null-space direction

s_σ : direction choice, zero in deadband

Entry offset and remaining angular error

Entry offset and remaining error: relative to the calibrated surface



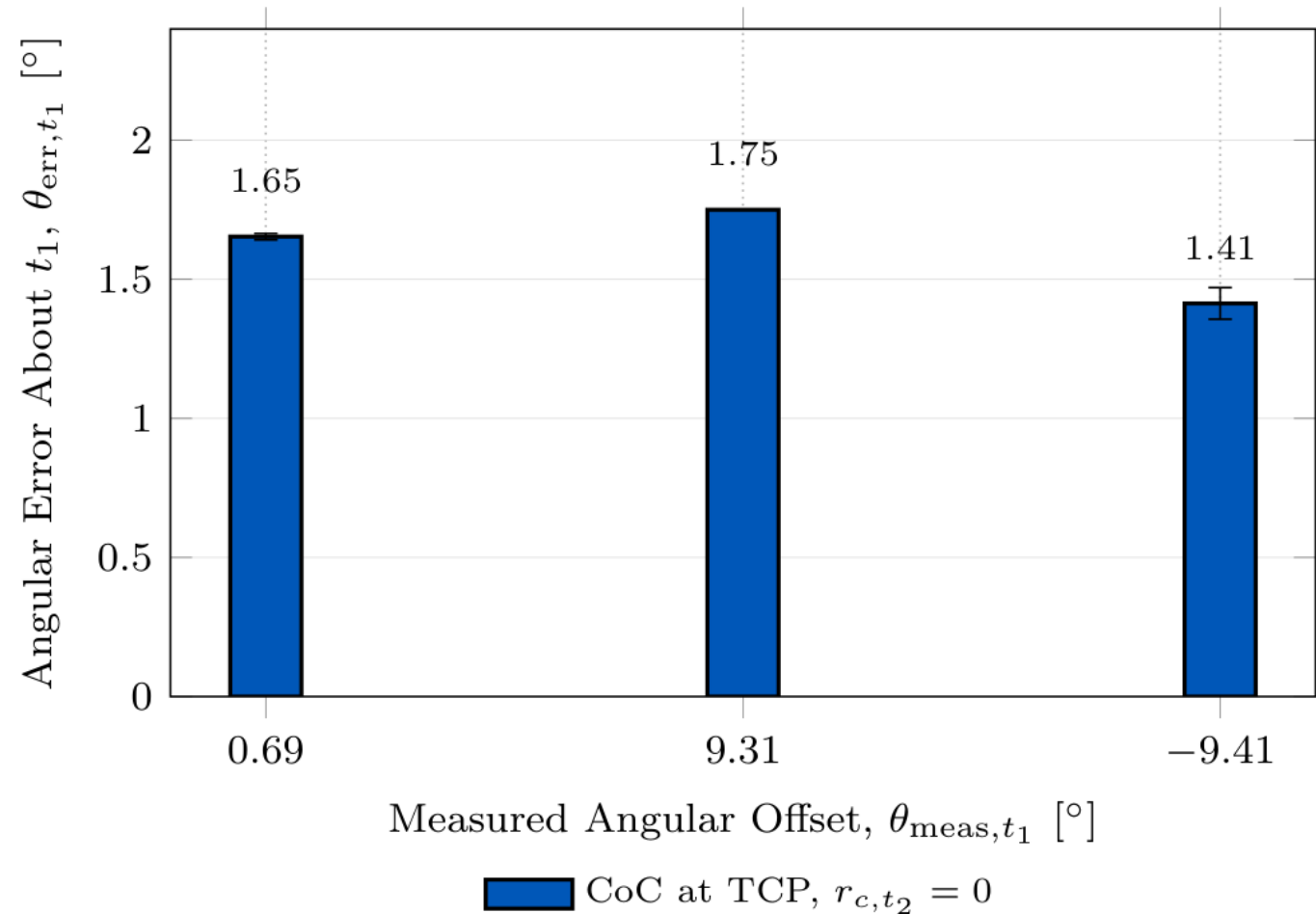
Plotted error: component about the first surface tangent

Angular error with CoC at the TCP

CoC at the TCP

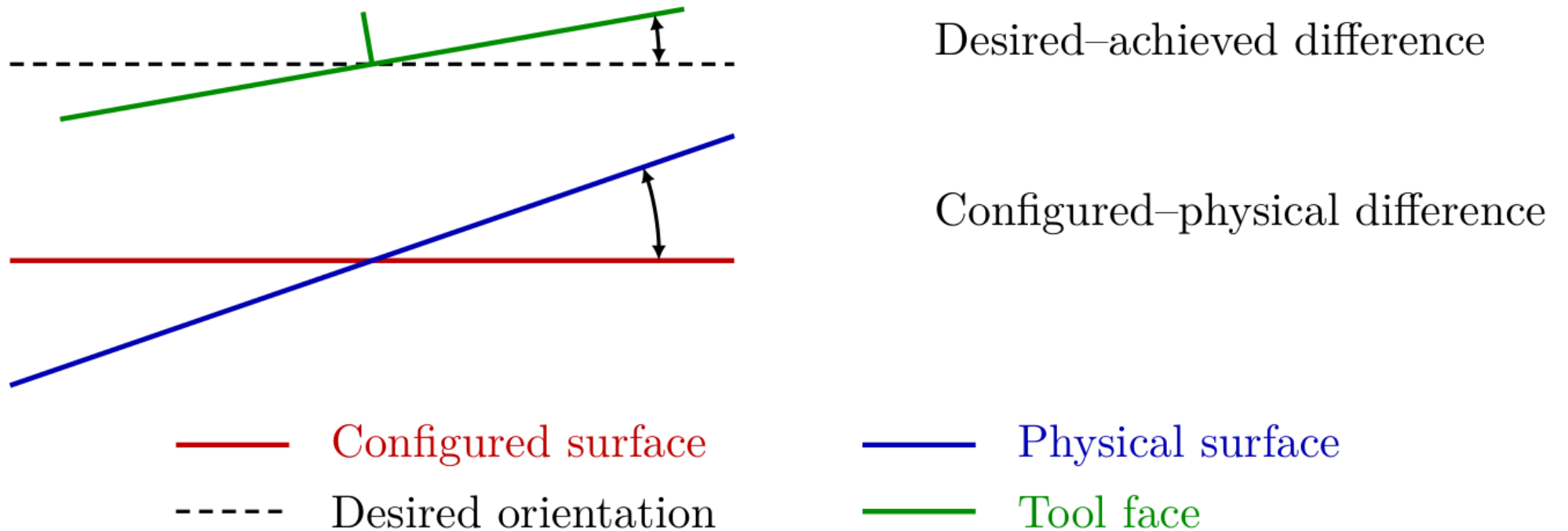
Small angular errors remain after contact

Reduced starting tilt
Residual angular error



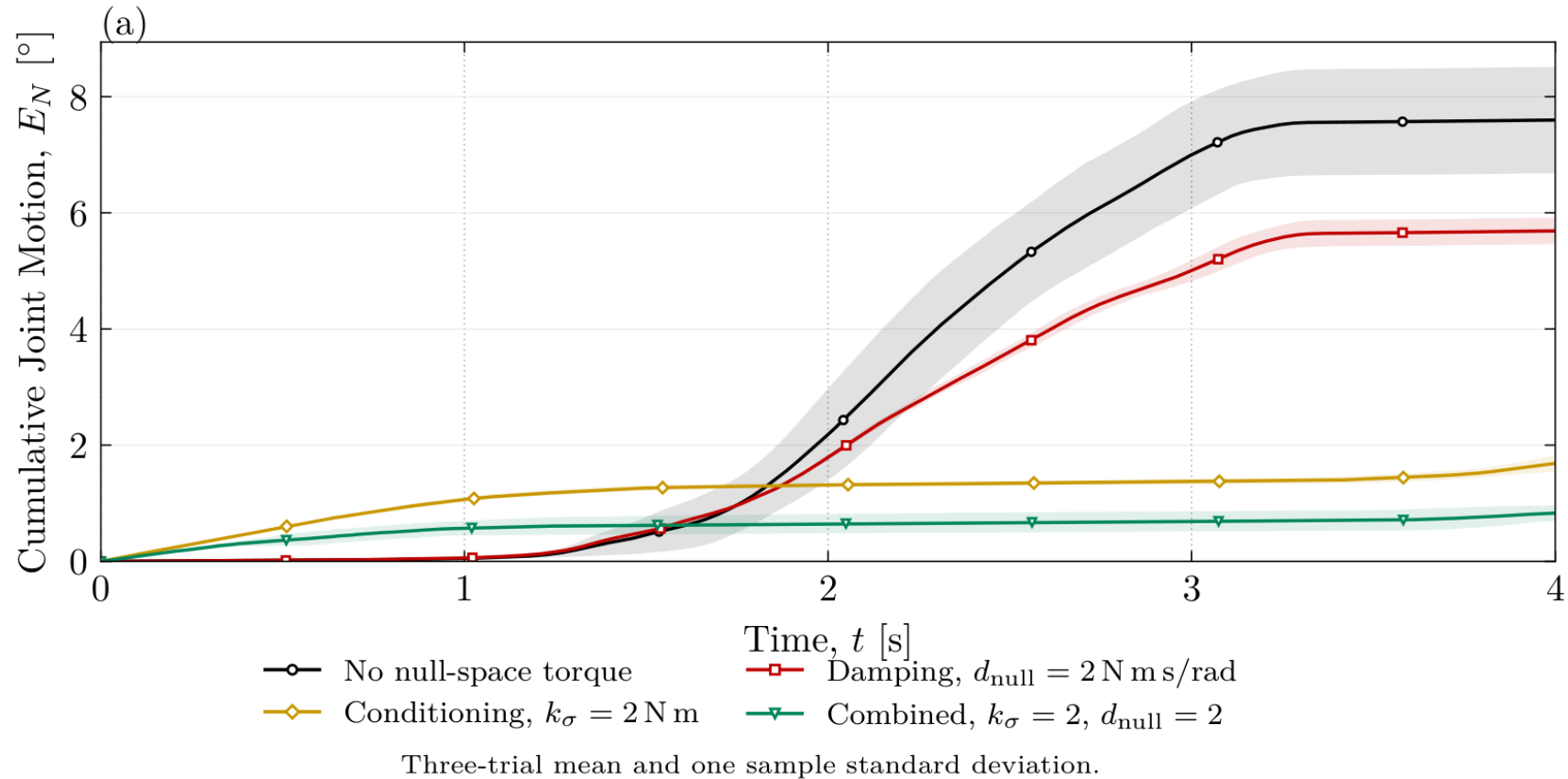
Sources of angular offset

Starting tilt: surface-calibration + tool-orientation differences



Both differences contribute to the entry angle

Damping reduces cumulative joint motion



≈25% less cumulative motion with damping alone

≈50% less cumulative motion with damping added to conditioning